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Beyond depression: correlates of well-being in young adult survivors of childhood cancers

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Abstract

Purpose—This study investigated the correlates of well-being with psychosocial and clinical factors in young adult childhood cancer survivors (YACCS) above and beyond depressive symptoms.

Methods—Participants were from the Project Forward Cohort, a population-based study of young adult survivors of childhood cancers. Participants (n = 1166, M_{age} = 25.1 years) were recruited through the Los Angeles Cancer Surveillance Program (Cancer Registry covering Los Angeles County). A majority received a diagnosis of leukemia (36.1%) or lymphoma (21.7%). Participants completed self-reported questionnaires at one timepoint. Multiple regression analyses were performed with well-being as the outcome variable and psychosocial and clinical variables (social support, sense of adulthood, posttraumatic growth, treatment intensity, and self-rated health) as the independent variables. Covariates included demographics (age, gender, relationship status, race/ethnicity) and depressive symptoms.

Results—In the multivariable model, posttraumatic growth, social support, sense of adulthood, and self-rated health were significantly associated with well-being (all ps < .05), when controlling for depressive symptoms. Treatment intensity and years since diagnosis were not significantly associated with well-being, when controlling for depressive symptoms.

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Ethics approval This study was approved by the IRB of University of Southern California (HS-14-00817) and the Committee for the Protection of Human Subjects (State of California) (CPHS protocol ID: 14-10-1755).

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent for publication Informed consent to publish was obtained from all individual participants included in the study.

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Conclusions—There are unique correlates of well-being above and beyond depressive symptoms among YACCS. This finding illuminates individual differences that may be associated with well-being and provides targets for intervention.

Implications for Cancer Survivors—Psychosocial interventions and survivorship care for YACCS should consider the broad aspects of well-being, independent of depressive symptoms.

Keywords

Well-being; Depressive symptoms; Posttraumatic growth; Childhood cancer; Adolescent and young adult

Introduction

Given improvements in cancer treatment, more than 80% of children and adolescents diagnosed with cancer have long-term survival [1, 2]. When childhood cancer survivors become young adults, they face unique challenges due to the interaction between the cancer experience and developmental tasks [3]. For example, taking time off from school or work creates significant delays in achieving adult milestones. Interruptions during their developmental trajectory may exacerbate psychological distress for some survivors [4]. Physical and social sequelae of cancer treatment that manifest in this life stage can significantly impact well-being [5].

Mental health is more than simply the presence or absence of distress or mental illness [6]. The dual continua model defines mental health as two distinct but linked constructs. Mental illness refers to the presence of psychiatric disorder, while mental health is defined as the presence of emotional, psychological, and social well-being [7–9]. This holistic perspective of well-being measures the extent to which an individual has an overall positive functioning in life (i.e., flourishing) [10].

Empirical evidence suggests that positive mental health indicators have protective associations with numerous health outcomes, above and beyond traditional indicators of distress (e.g., depressive symptoms) [11, 12]. For example, emotional well-being is associated with lower sleep disturbance, fatigue, and depressive symptoms [13]. Depressive symptoms are a major component of mental health, and are associated with poor physical, emotional, and cognitive functioning [14, 15]. Adolescent and young adult cancer survivors report poor mental health twice as often as other non-cancer controls [16]. Childhood cancer survivors experience significantly higher emotional distress and depression compared to control groups [17–20]. To date, research concerning the two distinct components of mental health are still underexplored in most prior cancer survivorship research [21, 22].

Previous studies report that psychosocial factors are positively related to various indices of well-being [13]. Research also shows that cancer-related clinical variables are associated with well-being. In one study of 194 young adult childhood cancer survivors, for instance, those who received a cancer diagnosis during adolescence and those with the highest levels of treatment intensity were at risk for psychological distress, adversely impacting well-being and quality of life [23]. Prior studies also report that the number of health problems reported

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by the survivors is negatively associated with overall well-being [24, 25]. These findings indicate the need for a global assessment of how perceived health may impact well-being even after controlling for depressive symptoms [26].

The association between well-being and psychosocial and clinical factors is limited to studies with relatively small sample sizes and/or a focus on older cancer survivors [13]. Thus, the aim of the current study was to use a population-based sample to investigate psychosocial (e.g., cancer-related posttraumatic growth, social support, sense of adulthood) and clinical (e.g., treatment intensity, years since diagnosis, and self-rated health) factors that are associated with well-being above and beyond depressive symptoms in analyses controlling for demographic covariates, such as age, gender, relationship status, and race/ethnicity. Based on the previous literature, we hypothesized that greater treatment intensity and lower self-rated health would be associated with lower well-being after controlling for depressive symptoms. We also hypothesized that higher cancer-related posttraumatic growth, sense of adulthood, and social support would be associated with higher well-being, after controlling for depressive symptoms and demographic covariates.

Methods

Participants and procedures

Data were collected as part of a larger population-based study, Project Forward 2, which investigated risk and protective factors of receiving cancer-related follow-up care among ethnically diverse adolescent and young adult childhood cancer survivors [27]. Participants were identified through the Los Angeles Cancer Surveillance Program, the Surveillance, Epidemiology, and End Results (SEER) Cancer Registry covering Los Angeles County (LAC). Eligibility criteria were as follows: the current age of 18 to 39 years old, a cancer diagnosis (stage 1 for brain cancer, stage 2 for others) in LAC between the years of 1996 and 2010, and currently 5 or more years after diagnosis. Identified participants were invited through mailings. Eligible and interested individuals were provided with a study questionnaire through mail, internet, or phone. The study procedures were approved by the Institutional Review Board at the University of Southern California.

Measures

Well-being—The Mental Health Continuum – Short-Form Scale (MHC-SF) was used to assess the psychological, emotional, and social well-being during the past month [7]. The participants were asked to respond how often they experienced a certain way, for example, “interested in life,” “you are good at managing the responsibilities of your daily life,” “our society is a good place,” or “people are basically good.” Response ranged from 0 = “Never” to 5 = “Every day.” The scale demonstrated good internal consistency in this sample (α = .94).

Treatment intensity—Treatment intensity was estimated using a method modeled after the Intensity of Treatment Rating Scale 2.0 (ITR-2) [28]. This scale incorporates cancer registry data and the survivors’ self-reported treatment and relapse information to categorize treatments into four levels of intensity: 1 = “Least intensive” (e.g., surgery only), 2 =

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"Moderately intensive" (e.g., chemotherapy or radiation), 3 = "Very intensive" (e.g., two or more treatment modalities), and 4 = "Most intensive" (e.g., relapse protocols).

Self-rated health—Self-rated health was assessed using one item from the SF-36: "How would you rate your general health overall?" [29]. It was scored using a 5-point Likert scale ranging from 0 = "Poor" to 4 = "Excellent."

Cancer-related posttraumatic growth—Eleven items were adapted from the Post-Traumatic Growth Inventory (PTGI) to measure personal growth stemming from the cancer experience [30]. Items assessed change experienced after cancer in "appreciation for the value of my own life" and "my own inner strength." Whereas the original scale measured the degree of positive change, the adapted scale measures both negative and positive changes. These items were scored using a 5-point scale ranging from 0 = "Highly negative change," 1 = "Somewhat negative change," 2 = "No change," 3 = "Somewhat positive change," to 4 = "Highly positive change." A mean posttraumatic growth score was calculated, with higher scores indicating more positive change in posttraumatic growth (Cronbach's α = .89).

Social support—A single item was used to assess perceived social support: "How many people do you have that you can count on for help when you need them?" This item was scored using a 3-point scale ranging from 0 = "None," 1 = "One to two," to 2 = "Three or more" [31].

Sense of adulthood—Sense of adulthood was assessed using the item "Do you feel like you have reached adulthood?" which was adapted from the emerging adulthood questionnaire [32]. This item was scored using a 3-point scale ranging from 0 = "No," 1 = "In some ways yes, in some ways no," to 2 = "Yes."

Depressive symptoms—Depressive symptoms were measured using the 20-item Center for Epidemiological Studies Depression Scale (CES-D) [33]. Participants indicated how often they experienced symptoms (e.g., sleep disruption, feeling of loneliness, guilt, and worthlessness, depressed mood, loss of appetite) during the previous week on a 4-point scale ranging from 0 = "Rarely or none of the time" (less than 1 day) to 3 = "Most or all of the time" (5–7 days). The total score was computed, and higher scores represented higher levels of depressive symptoms. (Cronbach's α = .80).

Demographic and clinical factors—Information on gender, race/ethnicity, relationship status, education attainment, employment status, age at diagnosis, and cancer type (e.g., skin, brain/central nervous system, endocrine, lymphoma) was obtained using both the SEER registry and self-reported questionnaire data. Current age, race/ethnicity, cancer type, and age at diagnosis were collected from the cancer registry.

Statistical analyses

Pearson correlations were conducted to assess the bivariate associations of psychosocial and clinical variables and well-being and depressive symptoms. Well-being was modeled using two multivariable linear regressions, one without depressive symptoms and one with depressive symptoms. In both models, clinical and psychosocial predictor variables

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were entered simultaneously. Data analyses were conducted using SAS statistical software (version 9.4; SAS Institute, Inc., Cary, NC).

Results

Participants

Of the 2788 eligible cases, 196 were deemed ineligible (e.g., deceased, too ill, or incompetent) and 1426 were excluded (e.g., declined to participate, gatekeeper refusal). The overall response rate was 44.9% and the participation rate was 64%. A total of 1166 childhood cancer survivors responded to the questionnaire. The final sample consisted of 1106 young adult survivors of childhood cancer; sixty participants who reported receiving cancer treatment within the last 2 years were not included in the analyses. Responder analyses were conducted using the demographic and clinical variables available from the registry. No differences were observed between non-responders and responders in age at diagnosis, years since diagnosis, current age, cancer type, and stage of disease [27].

The mean age was 26.2 (4.9) years old, and 49.2% identified as male and 50.8% as female (Table 1). The mean age at diagnosis was 11.6 (5.4) years. Half of the sample (51.1%) identified as Hispanic, 29.3% as White, 9.7% as Asian, and 9.5% as other. A majority was single (68.9%) and attained at least some college education (76%). More than 60% of the participants received very to most intensive treatment. One-third (33.66%) reported CES-D scores above 16 clinical cutoff point, indicating risk for clinical depression. Half of the sample (56.2%) reported well-being scores that indicated flourishing and 6.6% reported well-being scores that indicated languishing.

Main outcomes

Bivariate correlations between study variables are presented in Table 2. We observed the strongest inverse relationship between well-being and depressive symptoms ($r = -.67$, $p < .001$). The remaining variables, including cancer-related posttraumatic growth, social support, sense of adulthood, and self-rated health, were significantly and positively correlated with well-being with r's ranging from .07 to .50. Treatment intensity and years since diagnosis were not associated with well-being.

In analysis controlling for demographic covariates (i.e., race/ethnicity, age, gender, relationship status), cancer-related posttraumatic growth ($\beta = .37$), social support ($\beta = .16$), sense of adulthood ($\beta = .15$), and self-rated health ($\beta = .30$) were associated with well-being (all ps < .001) (Table 3). After controlling for depressive symptoms, these associations were attenuated. Depressive symptoms had the largest contribution to the model ($\beta = -.44$), but cancer-related posttraumatic growth ($\beta = .26$), social support ($\beta = .10$), sense of adulthood ($\beta = .08$), and self-rated health ($\beta = .18$) remained significant. Race/ethnicity was included as a covariate and Hispanic/Latinx origin remained significant after controlling for depressive symptoms ($\beta = -.07$, $p < .01$).

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Discussion

The present study examined associations between clinical and psychosocial factors with well-being, above and beyond depressive symptoms among young adult survivors of childhood cancer. Though depressive symptoms are a well-known correlate of well-being among young adults [34], our analyses suggest that posttraumatic growth, social support, self-rated health, and sense of adulthood are associated with well-being, above and beyond depressive symptoms. Contrary to hypotheses, clinical factors, such as treatment intensity, years since diagnosis, gender, and age, were not associated with well-being. Hispanic cancer survivors reported lower well-being compared to non-Hispanic survivors before and after controlling for depressive symptoms. This is in line with previous findings in which Hispanic childhood cancer survivors scored lower in quality of life and higher in depressive symptoms when compared to non-Hispanic survivors [21].

The results support the findings of previous studies showing that posttraumatic growth, sense of adulthood, and social support are positively related to indices of eudaimonic well-being among older adult cancer survivors [35, 36] and extend these findings to a younger population of cancer survivors. Despite the suggested impact of severity of treatment and years since diagnosis, we did not observe significant associations here, perhaps because their relative contribution to current well-being is outweighed by other more proximal factors such as self-rated health. This is an encouraging finding that indicates unmodifiable factors (such as prior treatment) are not the strongest determinants of well-being in cancer survivors.

Depressive symptoms and well-being are two important and related constructs. Our findings suggest that future studies should consider both depressive symptoms and well-being. To promote well-being directly, in addition to merely treating or preventing depressive symptoms, a range of interventions could be employed to target the various psychosocial factors identified in this study [37–40]. For example, we found that self-identifying as an adult was related to well-being. Because young cancer survivors may have delays in life milestone achievement, fostering skills and perceptions of achieving a sense of adulthood may be particularly impactful among young cancer survivors struggling with developing into adults while subsequently dealing with the aftermath of the cancer experience.

The cancer experience is one that often includes significant social isolation, and interventions that focus on social support among this population may lead to additional benefits if they fit the particular support needs of younger survivors [41, 42]. Other opportunities may include enhancing peer survivorship support groups, given our finding that social support was significantly associated with well-being. With advances in the number of platforms facilitating online social connection, numerous options are available to increase the accessibility of survivor social support. Some apps are dedicated to connecting survivors with similar experiences [43]. Survivorship clinics might leverage this by creating hybrid online and in-person communities to connect local survivors and facilitate peer social support [44]. Recent evidence also suggests that providing support for others can increase perceptions of social support among young adult cancer survivors [45, 46]. Short online interventions designed to elicit acts of support, therefore, may be an alternative approach

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to improving well-being in this population. Both posttraumatic growth and perceptions of health rely on shifts in meaning and refocused attention on the cancer experience. There are several innovative interventions, some based in narrative work that could be utilized with this population to assist cancer survivors achieve more optimal well-being [47, 48].

The strengths of the present study include the large, ethnically diverse, population-based sample of young adult survivors of childhood cancers, utilization of cancer registry data, and measurement of multiple facets of psychosocial and clinical variables and well-being. Nevertheless, conclusions are tempered due to the data being cross-sectional, such that causal relationships cannot be inferred. A prospective design that establishes temporal precedence between well-being and changes across time in psychosocial factors is recommended.

This study reports unique correlates of well-being among a sample of young adult survivors of childhood cancer. Our data indicate that several psychosocial factors are significantly associated with well-being above and beyond depressive symptoms. Psychosocial interventions and survivorship care for YACCS should consider the broad aspects of well-being, independent of the avoidance/prevention of depressive symptoms, to best increase the likelihood of flourishing in life. Positive psychology interventions provide unique opportunities for addressing well-being and can be considered to support the psychosocial and clinical care for young adult childhood cancer survivors, including fostering optimal transition to adulthood and life-long well-being.

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Data availability

The data underlying this article cannot be shared publicly due to privacy restrictions of individuals that participated in the study. Aggregated, deidentified data may be shared on reasonable request to the corresponding author.

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Table 1

Descriptive statistics of childhood cancer survivors' clinical, psychosocial, and demographic factors (*n*=1106; diagnosed in 1996–2010; Los Angeles County)

Variable	Mean (SD), range or *N* (weighted %)
Sex	
Male	544 (49.2)
Female	562 (50.8)
Age at survey completion (years)	26.2 (4.9), 18–41
Age group at survey completion (years)	
18–20	131 (11.8)
21–25	422 (38.2)
26–30	339 (30.6)
31–39	214 (19.4)
Age at diagnosis	11.6 (5.4), 1–19
Years since diagnosis	14.52 (4.36)
Race/ethnicity	
White	324 (29.3)
Hispanic/Latino	570 (51.5)
Asian	107 (9.7)
Other	105 (9.5)
Cancer type	
Leukemia	392 (36.1)
Lymphoma	240 (21.7)
Brain and other nervous system	169 (15.2)
Endocrine system	60 (5.1)
Bones and joints	56 (5.0)
Skin	41 (3.5)
Genital system	56 (5.2)
Other[a]	92 (8.2)
Treatment intensity[b]	
1 (least intensive)	69 (6.0)
2 (moderately intensive)	344 (30.9)
3 (very intensive)	544 (49.9)
4 (most intensive)	149 (13.3)
Relationship status	
Single	751 (68.9)
Married	201 (18.4)
Living with a partner	111 (10.2)
Divorced/Separated	26 (2.4)
Widowed	1 (0.1)
Education	
Grade school (<8 years)	5 (0.5)

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Variable	Mean (SD), range or *N* (weighted %)
Some high school (<12 years)	53 (4.9)
High school graduate or GED (12 years)	205 (18.8)
Some college, vocational or training school	417 (38.2)
Associate degree	95 (8.7)
College graduate	241 (22.1)
Post graduate education	75 (6.9)
Well-being	47.3 (15.0), 4–70
Depressive symptoms	13.88 (10.9), 0–58
Cancer-related posttraumatic growth	3.93 (.7), 1.3–5
Social support	
None	30 (2.7)
One or two	377 (34.5)
Three or more	686 (62.8)
Self-rated health[c]	2.34 (1.0), 0–4
Sense of adulthood (yes)	586 (54.0)

[a]Oral cavity and pharynx, digestive system, respiratory system, soft tissue including heart, urinary system, eye and orbit, miscellaneous

[b]Base on the Intensity of Treatment Rating (ITR-3), Division of Oncology, Center for Childhood Cancer Research, The Children's Hospital of Philadelphia

[c]Higher scores indicate better health

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Table 2

Bivariate correlations between all variables

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) Posttraumatic growth							
(2) Social support	.21***						
(3) Adulthood	.18***	.08*					
(4) Self-rated health	.25***	.21***	.07*				
(5) Treatment intensity	.07*	−.03	.02	−.12***			
(6) Years since dx	−.07*	−.01	.08**	−.07*	−.06		
(7) Depressive symptoms	−.38***	−.29***	−.26***	−.41***	.02	−.01	
(8) Well-being	.50***	.33***	.27***	.45***	−.05	−.03	−.67***

* $p < .05$,

** $p < .01$,

*** $p < .001$

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Table 3

Regression analyses for psychosocial and clinical correlates of well-being (n = 995, 931)

	Model without depressive symptoms		Model with depressive symptoms	
	b **(SE *b*)**	***β***	***b*** **(SE *b*)**	***β***
Cancer-related posttraumatic growth	8.08 (.57)	.37***	5.64 (.54)	.26***
Social support	4.49 (.70)	.16***	2.86 (.65)	.10***
Sense of adulthood	4.50 (.79)	.15***	2.51 (.72)	.08***
Self-rated health	4.46 (.39)	.30***	2.63 (.37)	.18***
Treatment intensity	−0.30 (.47)	−.02	−0.56 (.43)	−.03
Years since diagnosis	−0.01 (.09)	−.00	−0.04 (.08)	−.01
Age at survey completion	−0.02 (.09)	−.01	−0.03 (.08)	−.01
Female (vs. male)	0.55 (.73)	.02	1.04 (.66)	.03
Single (vs. not single)	−1.48 (.90)	−.05	−0.90 (.81)	−.03
Race/ethnicity (vs. Non-Hispanic White)				
Hispanic	−2.76 (.88)	−.09**	−2.15 (.80)	−.07**
Asian	−2.86 (1.34)	−.06*	−1.86 (1.20)	−.04
Other	−2.65 (1.37)	−.05	−1.81 (1.24)	−.04
Depressive symptoms			−0.62 (.04)	−.44***
F value	61.17***		94.43***	
Adjusted R^2	.42		.57	

All analyses controlled for shared variance of well-being, depressive symptoms, and covariates (i.e., age at survey completion, gender, relationship status, race/ethnicity)

*$p < .05$,

**$p < .01$,

***$p < .001$